INCH-POUND

MIL-PRF-49471/5A(CR)
30 November 2000
Superseding
MIL-PRF-49471/5(ER)
2 June 1995

PERFORMANCE SPECIFICATION SHEET

BATTERY, NON-RECHARGEABLE, LITHIUM SULFUR DIOXIDE,
HIGH PERFORMANCE, BA-5847/U

This specification sheet is approved for use within the Communications Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification and MIL-PRF-49471(CR).

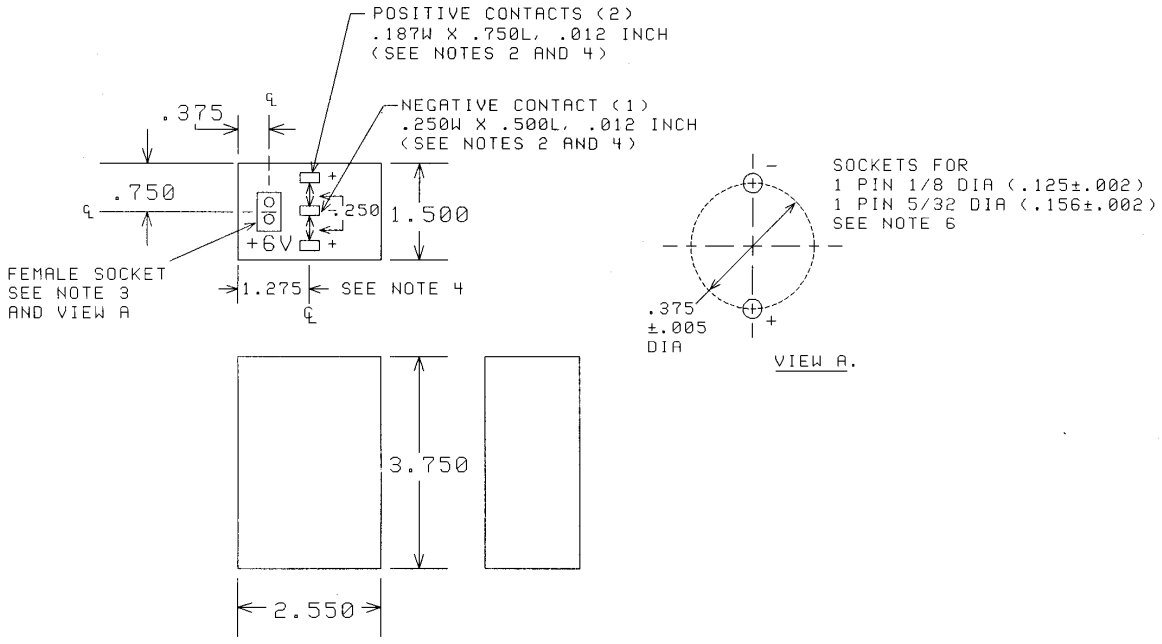

Figure 1. Battery dimensions and terminal markings.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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NOTES:

1. All dimensions are in inches. The tolerance shall be ±0.031 inch unless otherwise specified.

2. The dual connectors shall be wired in a parallel 6 volt circuit.

3. Type IV socket:

    a. Pin circle center shall be on the battery center line.

    b. Minimum depth of the socket well shall be 5/8 inch.

    c. Shall float 1/32 inch minimum in all directions from the pin circle center.

    d. Positive terminal shall be the 5/32 inch dia. Socket (see Figure 1, View A).

4. Flat contacts:

    a. The center negative contact shall be on both battery center lines.

    b. Shall be supported and mounted so that the top surface shall not protrude above the outside surface of the battery jacket.

    c. Shall not be more than 1/8 inch below the outside surface of the battery jacket initially, during, and after contact is made with mating contact springs.

    d. Shall not have a lateral displacement of more than 1/32 inch during mating with contact springs.

5. Terminal polarity and voltage shall be as shown in Figure 1.

6. Insertion Force: 12 pounds MAX. Withdrawal Force: 3 pounds MIN.

7. Socket requirements to be certified in the First Article Test Report.

REQUIREMENTS:

Dimensions and configuration: See Figure 1.

Maximum voltage: 6.10 volts.

Cut-off voltage for capacity: 4.00 volts.

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Dual Terminals: Two (2) hole female socket per Figure 1, View A; three flat contacts as per Figure 1.

Weight (maximum): 0.6 pound (272 grams).

Battery short circuit test: Applicable.

Charge protection: Applicable.

High temperature protection: Applicable.

Complete discharge device: Applicable.

State of Charge Indicator: Not applicable.

Capacity tests: When the battery is tested in accordance with the methods of examination and tests of this specification, the minimum capacity test requirements shall be as specified below:

Capacity Test	Minimum Capacity Requirements in Hours to 4.00 Volts
I	3.0
L	1.6
H	5.6
HT	5.4
LT	1.4
IT	2.8
LR1	48.0

Initial voltage delay: During the I, H, HT, and IT discharge tests initial voltages below 4.00 volts cannot exceed a 5.0 second duration. During the L and LT discharge tests initial voltage below 3.80 volts cannot exceed a 60.0 second duration.

METHODS OF EXAMINATION AND TESTS:

Capacity tests:

(1) Storage:

Details on storage conditions for all capacity tests are specified in the basic specification.

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(2) Discharge:

Cells (forced discharge): Two cells shall be discharged at 2.0 amperes at 70±5°F to 2.0 volts. Each discharged cell shall then be connected in series to one fresh cell. Each string shall be discharged at 2.0 amperes at 70±5°F for a minimum of 3.0 hours.

Batteries: For all capacity tests, except the LR1, H and HT, the batteries shall be discharged at 2.0 amperes constant current to cut-off volts. For the H and HT capacity tests the batteries shall be discharged through 2.0 amperes constant current for five minutes followed by open circuit for five minutes. This cycle shall be repeated until the closed-circuit voltage reaches cut-off volts. For the LR1 test, batteries shall be discharged at 135 milliamperes to 4.5 volts, then at 1.1 watts to 1 volt.

Forced Discharge: After discharge to zero volts on the I capacity test, batteries shall be force discharged at 2.0 amperes for a minimum of 5 minutes.

Overload test: Batteries shall be discharged at 4 amperes.

Abuse test predischarge: Designated samples shall be discharged at 2.0 amperes for 1.75 hours.

Abuse test, pulse discharge: Batteries shall be discharged at 2.5 amperes for 1 minute followed by 1.8 amperes for 4 minutes, cycled continuously to cut-off voltage.

Closed circuit voltage test (battery and cell string): Load resistance of 2.0 ohms shall be used. Voltage shall be above 4.00 volts within 10 seconds.

Cell closed circuit voltage test: Load resistance of 1 ohm shall be used. Voltage shall be above 2.0 volts within 5 seconds.

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Army – CR

Preparing Activity:
Army - CR

(Project 6135-0417-05)

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